

Ab Initio Electron-Molecule Scattering Calculations for Methane Using UKRMol+ and ASTRA

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1 Abstract

Electron collisions with molecules are fundamental to many areas of physics and chemistry, including plasma processing, semiconductor manufacturing, and atmospheric chemistry. To model these processes, accurate electron-molecule scattering cross sections, which describe the probability of elastic and inelastic collision processes over a range of electron energies, are required. While cross sections can be measured experimentally, theoretical calculations provide insight into the underlying collision dynamics and supply data where measurements are difficult to obtain or unavailable. Specifically, in the low energy regime where the scattering electron interacts strongly with the molecular potential, accurately accounting for target-state polarization and electron correlation effects is essential for obtaining reliable scattering cross sections and meaningful agreement with experimental results.

In this project, low energy electron scattering calculations were conducted for methane (CH_4) using the UKRMol+ and ASTRA electron-molecule scattering packages. CH_4 was used as the primary benchmark species due to its relatively simple electronic structure and the availability of published experimental and theoretical cross sections for comparison. A series of calculations was performed to investigate how different physical and numerical approximations influenced the calculated results. By systematically varying the treatment of target-state polarization, electron correlation, active spaces, and scattering model parameters, the role each parameter played in determining the accuracy and convergence of the calculated integral elastic and inelastic cross sections was examined. The methodology developed for CH_4 will provide the foundation for future studies of more complex systems, such as tetrafluoromethane (CF_4), a PFAS gas used in plasma etching. Through this work, quantum scattering calculations with different degrees of electron correlation were conducted to predict electron-molecule scattering cross sections, which will enable the systematic production of scattering data of a wide range of molecules.

2 Introduction

Electron collisions with atoms and molecules play a fundamental role in a wide range of physical and chemical environments and have diverse applications in numerous fields including plasma processing, semiconductor manufacturing, atmospheric chemistry, and astrophysics. In these systems, electron-impact processes govern ionization, electronic excitation, dissociation, and attachment reactions that determine the chemical composition and transport properties of the medium. Accurate electron-molecule scattering cross sections are therefore essential inputs for plasma kinetics simulations, Monte Carlo transport calculations, collisional-radiative models, and the design and optimization of plasma-based technologies.

Particularly important in low-energy electron scattering modeling are two characteristic features of the cross section curve: the Ramsauer-Townsend (R-T) minimum and resonance peaks. The R-T minimum is a decrease in the scattering cross section that arises from perfect destructive interference between incident and distorted pathways. Resonance peaks occur when the incident electron is temporarily captured into a quasi-bound state of the target, forming a transient negative ion before autodetachment. These temporary anion states enhance the scattering probability and are often accompanied by vibrational distortion, electronic excitation, and sometimes even molecular dissociation. The energies, positions, and magnitudes of the R-T minimum and resonance peaks are governed by quantum mechanical effects including electron exchange, correlation, and target polarization.

Despite their importance, accurate theoretical calculations of electron-molecule scattering collisions remain computationally challenging because they require an accurate quantum mechanical description of both the target molecule and the scattering electron. The incoming electron interacts simultaneously with every electron and nucleus in the target molecule, giving rise to an $(N+1)$ -electron problem whose solution must properly account for electron exchange and correlation as well as target polarization effects. Since exact solutions of the many-electron Schrödinger equation are only possible for the simplest atomic systems, practical scattering calculations rely on a hierarchy of approximations in both the electronic structure and scattering formalisms.

Among the most successful *ab initio* methods for low-energy electron-molecule scattering are the close-coupling and R-matrix approaches implemented in codes such as UKRMol+ and ASTRA. Although the two codes employ different numerical strategies for solving the scattering problem, both begin by constructing accurate N -electron target wavefunctions using post-Hartree-Fock electronic structure methods before solving the corresponding $(N+1)$ -electron scattering equations. The quality of the target electronic structure is therefore one of the dominant factors governing the accuracy of the computed scattering cross sections and, in particular, the ability to model accurate R-T minima and resonance features. The following sections present the theoretical framework underlying scattering cross section calculations, beginning with the quantum mechanical description of molecular electronic structure and progressing through Hartree-Fock theory, electron correlation methods, configuration interaction, and the close-coupling scattering formalism used throughout this work.

2.1 Quantum Mechanical Description of Molecules

The electronic structure of a molecule is governed by the time-independent Schrödinger equation,

$$H|\Phi\rangle = E|\Phi\rangle \quad (1)$$

where H is the Hamiltonian operator, $|\Phi\rangle$ is the many-electron wavefunction, and E is the corresponding energy eigenvalue. The Hamiltonian contains all kinetic and electrostatic interactions and completely determines the quantum mechanical behavior of the system. For instance, for a molecule containing N electrons and M nuclei, the non-relativistic Hamiltonian in atomic units is

$$H = -\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{A=1}^M \frac{1}{2M_A} \nabla_A^2 - \sum_{i=1}^N \sum_{A=1}^M \frac{Z_A}{r_{iA}} + \sum_{i=1}^N \sum_{j>t}^N \frac{1}{r_{ij}} + \sum_{A=1}^M \sum_{B>A}^M \frac{Z_A Z_B}{R_{AB}}. \quad (2)$$

The first and second terms describe the kinetic energies of the electrons and nuclei, the third term describes electron-nucleus Coulombic attraction, and the fourth and fifth terms describe electron-electron and nucleus-nucleus Coulombic repulsion, respectively.

Due to the electron-electron repulsion term in the Hamiltonian, solving the time-independent Schrödinger equation becomes difficult, especially for larger systems. This interaction term couples the motion of every electron to every other electron, making the many-electron Schrödinger equation non-separable. Consequently, analytical solutions exist only for one-electron systems such as hydrogen, while molecules containing multiple electrons require numerical approximations.

2.2 The Born-Oppenheimer Approximation

One of the most fundamental approximations is the Born-Oppenheimer approximation. This approximation exploits the large difference in mass between electrons and nuclei. Since nuclei are several orders of magnitude heavier than electrons, they move much more slowly. Electronic motion typically occurs on attosecond time scales, whereas nuclear vibrational motion occurs on femtosecond time scales. As a result, electrons can be regarded as responding almost instantaneously to changes in the nuclear positions. Under this approximation, the electronic and nuclear degrees of freedom are separated, allowing for the total molecular wavefunction to be expressed as

$$\Phi(\{r_i\}; \{R_A\}) = \Phi_{elec}(\{r_i\}; \{R_A\}) \Phi_{nucl}(\{R_A\}) \quad (3)$$

where Φ_{elec} describes the electronic motion in the field of fixed nuclei and Φ_{nucl} represents the nuclear wavefunction.

Substituting this factorization into the molecular Schrödinger equation and neglecting the coupling between electronic and nuclear motion results in the electronic Schrödinger equation. Thus, by using the Born-Oppenheimer approximation, the nuclei present in the system are treated as fixed point charges and the electronic wavefunction can be solved for independently. Since the present work focuses exclusively on electronic scattering processes and neglects vibrational and rotational excitation, the Born-Oppenheimer approximation holds and can be used to describe the electronic state of the target molecule.

2.3 Hartree-Fock Theory

Even after the application of the Born-Oppenheimer approximation, solving the electronic Schrödinger equation remains challenging because the wavefunction depends simultaneously on the coordinates of all electrons. However, if electron-electron interactions are neglected, the many-electron wavefunction can be approximated as a product of one-electron spin orbitals

$$\chi_1(x_1)\chi_2(x_2)\cdots\chi_N(x_N) \quad (4)$$

where x_i denotes the combined spatial and spin coordinates of electron i and $\chi_i(x)$ is a spin orbital. This product wavefunction, known as the Hartree product, assumes that each electron moves independently of every other electron. The corresponding energy (E) therefore is the sum of all individual orbital energies (ε),

$$E = \sum_i \varepsilon_i. \quad (5)$$

Although computationally convenient, the Hartree product fails to account for the indistinguishability of electrons. Exchanging the coordinates of any two electrons changes the wavefunction, violating the antisymmetry requirement imposed by the Pauli exclusion principle. Consequently, the Hartree product cannot represent a physically acceptable many-electron wavefunction. To satisfy the Pauli exclusion principle, the total electronic wavefunction must be antisymmetric with respect to the interchange of the spatial and spin coordinates of any two electrons. This antisymmetry requirement ensures that no two electrons may occupy the same spin orbital and introduces exchange effects. The simplest wavefunction that satisfies these conditions is the Slater determinant,

$$\Psi(x_1, x_2, \dots, x_N) = (N!)^{-1/2} \begin{vmatrix} \chi_i(x_1) & \chi_j(x_1) & \cdots & \chi_k(x_1) \\ \chi_i(x_2) & \chi_j(x_2) & \cdots & \chi_k(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_i(x_N) & \chi_j(x_N) & \cdots & \chi_k(x_N) \end{vmatrix} \quad (6)$$

where each column corresponds to a spin orbital and each row corresponds to the spatial and spin coordinates of a single electron. Interchanging the coordinates of any two electrons exchanges two rows of the matrix. Therefore, the Slater determinant changes sign whenever two electrons are exchanged, resulting in a wavefunction that satisfies the antisymmetry requirement. Furthermore, if two electrons were to occupy the same spin orbital, two columns of the matrix would become identical, causing the determinant to be zero. Thus, the Slater determinant satisfies the Pauli exclusion principle while providing a physically meaningful approximation to the many-electron wavefunction.

In addition to enforcing this antisymmetry, the Slater determinant introduces exchange interactions that correlate the motion of electrons with parallel spins. However, electrons with opposite spins remain uncorrelated within the single-determinant description. This single-determinant description forms the foundation of Hartree-Fock theory, which is the starting point for most *ab initio* electronic structure methods.

Following the Born-Oppenheimer approximation and the construction of an antisymmetric wavefunction, the electronic Schrödinger equation must still be resolved. Unfortunately,

the electron-electron Coloumb repulsion couples the motion of every electron to every other electron, making the exact many-electron problem analytically challenging. Hartree-Fock theory provides a systematic approximation to the solution by replacing the exact many-electron wavefunction with a single Slater determinant constructed from one-electron spin orbitals and determining optimal orbitals through the variational principle.

The Hartree-Fock equations are derived using the variational principle, which states that the expectation value of the Hamiltonian for any normalized trial wavefunction is always greater than or equal to the ground-state energy

$$E_0 \leq \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle}. \quad (7)$$

The optimal single-determinant wavefunction is therefore obtained by minimizing the total electronic energy with respect to the spin orbitals while maintaining orthonormality. This procedure yields the Hartree-Fock equations,

$$f(i)\chi(x_i) = e_i\chi(x_i) \quad (8)$$

where $f(i)$ is the Fock operator, e_i is the orbital energy, and χ_i is the corresponding spin orbital. The Fock operator may be expressed as

$$f(i) = -\frac{1}{2}\nabla_i^2 - \sum_{A=1}^M \frac{Z_A}{r_{iA}} + v^{HF}(i) \quad (9)$$

where the first two terms represent the one-electron kinetic energy and electron-nucleus attraction, while v_{HF} denotes the average electrostatic potential generated by all remaining electrons. Unlike the exact Hamiltonian, the Hartree-Fock approximation replaces the instantaneous electron-electron interaction with a mean field, allowing the many-electron problem to be reduced to a set of coupled one-electron equations.

Since the Fock operator depends on the occupied spin orbitals, which themselves are unknown, the Hartree-Fock equations must be solved through an iterative process. Beginning with an initial guess for the molecular orbitals, a Fock matrix is constructed and diagonalized to obtain an improved set of orbitals. This process is repeated until the input and output orbitals are self-consistent, giving rise to the self-consistent field (SCF) procedure.

In practice, the Hartree-Fock equations cannot be solved analytically for molecules and must instead be solved numerically by expanding each molecular orbital as a linear combination of a finite set of known atomic basis functions. This approach, known as the linear combination of atomic orbitals (LCAO) approximation, expresses the i th molecular orbital as

$$\psi_i(r) = \sum_{\mu} c_{\mu_i} \phi_{\mu}(r) \quad (10)$$

where ϕ_{μ} are atomic basis functions and c_{μ_i} are expansion coefficients. Rather than solving directly for the molecular orbitals, the Hartree-Fock procedure determines the set of coefficients that produce the lowest-energy electronic wavefunction.

Several types of basis functions have been developed for electronic structure calculations. Gaussian-type orbitals (GTOs) are commonly employed. Although a single Gaussian function does not reproduce the correct exponential decay of an atomic orbital, GTOs permit the

molecular integrals required in Hartree-Fock theory to be evaluated efficiently. The product of two Gaussian functions is itself another Gaussian. As a result, the one- and two- electron integrals required to construct the Fock matrix can be evaluated analytically, leading to reductions in computational cost.

Substituting the LCAO expansion into the Hartree-Fock equations produces the Roothaan-Hall equations,

$$FC = SC\epsilon \quad (11)$$

where F is the Fock matrix, S is the overlap matrix, C contains the molecular orbital coefficients, and ϵ is the diagonal matrix of orbital energies. Solving this generalized eigenvalue problem yields the occupied and virtual molecular orbitals that define the Hartree-Fock wavefunction and provide the starting point for all subsequent post-Hartree-Fock methods, including configuration interaction calculations.

2.4 Electron Correlation and Configuration Interaction

Although Hartree-Fock theory provides an excellent qualitative description of molecular electronic structure, it remains fundamentally limited because the wavefunction consists of a single Slater determinant. Consequently, it accounts for exchange effects exactly but neglects the dynamic correlation arising from the instantaneous motion of electrons with opposite spin. The difference between the exact non-relativistic electronic energy and the Hartree-Fock limit defines the electronic correlation energy,

$$E_{corr} = \epsilon_0 - E_0 \quad (12)$$

where ϵ_0 is the exact non-relativistic ground-state energy within the Born–Oppenheimer approximation and E_0 is the Hartree-Fock limit. The correlation energy therefore represents the additional stabilization obtained by allowing the motions of all electrons to be correlated. Electrons avoid one another because of Coulomb repulsion, producing correlated motion that lowers the total energy, compared to the energy predicted by the independent-particle approximation of Hartree-Fock theory.

To recover this missing correlation energy, post-Hartree-Fock methods expand the electronic wavefunction beyond a single Slater determinant. One of the most widely used approaches is configuration interaction (CI), which expresses the exact wavefunction as a linear combination of multiple electronic configurations,

$$|\Phi\rangle = c_0|\Phi_0\rangle + \sum_{ra} c_a^r |\Psi_a^r\rangle + \sum_{a<b, r<s} c_{ab}^{rs} |\Psi_{ab}^{rs}\rangle + \dots \quad (13)$$

The reference determinant, $|\Phi_0\rangle$, is typically defined as the Hartree-Fock ground state, while the remaining determinants correspond to excitations of electrons from occupied orbitals into virtual orbitals. Single excitations promote one electron, double excitations promote two electrons simultaneously, and higher-order excitations proceed analogously. By allowing the wavefunction to consist of multiple determinants rather than a single electronic configuration, CI provides the additional variational freedom required to describe electron correlation.

The CI coefficients are determined variationally by diagonalizing the electronic Hamiltonian in the basis of the selected Slater determinants. As the number of included determinants

increases, the calculated energy approaches the exact solution within the chosen basis set. In the limit that every possible determinant is included, the method becomes Full Configuration Interaction (FCI), which provides the exact solution to the electronic Schrödinger equation for a finite basis. Although FCI is exact, the number of determinants increases combinatorially according to

$$\binom{2K}{N} \quad (14)$$

where N is the number of electrons and $2K$ is the number of spin orbitals. Consequently, FCI calculations often require millions or billions of determinants, making them computationally infeasible for most molecules of chemical interest. Practical scattering calculations therefore employ truncated or active-space CI methods that recover the most important electron correlation effects while remaining computationally tractable.

2.5 Complete Active Space Configuration Interaction (CASCI)

To obtain an accurate description of electron correlation while maintaining computational feasibility, both UKRMol+ and ASTRA employ Complete Active Space Configuration Interaction (CASCI). Rather than allowing excitations among every molecular orbital, CASCI partitions the orbital space into three regions: frozen orbitals, active orbitals, and virtual orbitals.

Core orbitals are treated as doubly occupied throughout the calculation, while high-energy virtual orbitals remain unoccupied. Only electrons within the active orbitals are allowed to undergo every possible electronic excitation. The resulting CASCI wavefunction is therefore written as

$$|\Psi_{MCSCF}\rangle = \sum_I c_I |\Psi_I\rangle \quad (15)$$

where each determinant corresponds to a different occupation pattern within the active space.

The size of the active space determines the balance between computational expense and accuracy. Increasing the number of active orbitals improves the description of electron correlation and target polarization but also increases the dimension of the CI Hamiltonian drastically. Thus, careful selection of the active space is essential for obtaining accurate scattering calculations within reasonable computational cost.

The resulting target states serve as the electronic basis for the subsequent electron-molecule scattering calculation. Since low-energy collisions may excite the target into numerous electronic states, multiple CASCI target states are generally required to accurately represent polarization effects during collision.

2.6 Electron-Molecule Scattering Theory

Once the electronic structure of the target molecule has been determined, the scattering calculation considers the interaction between the incident electron and the target. The collision therefore becomes an $(N+1)$ -electron problem whose exact wavefunction satisfies the many-electron Schrödinger equation.

One of the most successful approaches for describing low-energy electron–molecule collisions is the close-coupling approximation. Rather than solving directly for the complete $((N+1))$ -electron wavefunction, the total scattering wavefunction is expanded as a linear combination of the target electronic states coupled to the motion of the projectile electron,

$$\Psi^\Gamma(x_1, \dots, x_N, x) = A \sum_i \sum_j a_{i,j}^\Gamma \Phi_i^\Gamma(x_1, \dots, x_N, \hat{x}) \frac{1}{r} F_{E,i,j}(r) + \sum_k b_k^\Gamma \chi_k^\Gamma(x_1, \dots, x_N, x_{N+1}) \quad (16)$$

where Γ denotes the total spin and spatial symmetry of the scattering system and A is the antisymmetrization operator, which ensures that the scattering electron remains indistinguishable from the target electrons. The channel functions Φ_i^Γ are constructed by coupling the N -electron target states with the angular and spin coordinates of the scattering electron, while the radial functions $F_{E,i,j}(r)$ describe the motion of the scattered electron. The square-integrable configurations, χ_k^Γ , provide additional description of short-range electron correlation and target polarization within the interaction region, where the projectile perturbs the target molecule’s electronic structure.

The coefficients $a_{i,j}^\Gamma$ and b_k^Γ are determined by diagonalizing the $(N+1)$ -electron Hamiltonian separately for each total spin and spatial symmetry. Exploiting molecular symmetry substantially reduces the computational cost because the Hamiltonian becomes block diagonal, allowing each irreducible representation to be treated independently. In R-matrix theory, these coefficients correspond to the eigenvectors obtained by diagonalizing the inner-region Hamiltonian matrix.

The scattering problem may be formulated rigorously using the Lippmann-Schwinger equation:

$$\langle k_f \Phi_f | T | k_i \Phi_i \rangle = \langle k_f \Phi_f | V | \Psi_i^+ \rangle \quad (17)$$

where $\langle k_i \Phi_i$ and $\langle k_f \Phi_f$ represent the initial and final asymptotic scattering states, respectively, with k_i and k_f denoting the linear momenta of the projectile electron and Φ_i and Φ_f denoting the corresponding target states. V denotes the interaction potential, $|\Psi_i^+\rangle$ is the exact solution of the scattering problem, and T is the transition operator. The transition matrix elements describe the probability amplitudes for transitions between the initial and final scattering channels.

Both UKRMol+ and ASTRA solve this scattering problem using close-coupling methods, although the codes employ different numerical methods. In UKRMol+, the scattering equations are solved within the R-Matrix formalism by partitioning space into an inner region, where electron exchange and correlation are explicitly included, and an outer region, where exchange becomes negligible and only long-range multipole interaction remains. Target states are computed using Hartree-Fock theory and target wavefunctions are represented using CASCI. ASTRA solves the coupled scattering equations differently by using transition density matrix close coupling (TDMCC) formalism without introducing an R-Matrix boundary. The Lippmann-Schwinger equation is solved through direct continuum-basis expansion of the scattering wavefunction. While ASTRA similarly uses Hartree-Fock theory and CASCI, the code additionally optimizes target states through state averaging and computes exchange terms exactly. Despite these algorithmic differences, both approaches solve the same underlying quantum mechanical scattering problem to produce scattering cross section observables.

The scattering calculation yields the scattering (S) matrix, which characterizes the collision. The transition matrix is related to the scattering matrix through

$$T = S - I \quad (18)$$

where I is the identity matrix.

The primary observable computed in this work is the integral cross section (ICS), which represents the total probability that an incident electron scatters into a specified final electronic state after integrating over all scattering angles. The state-to-state integral cross section is obtained directly from the S-matrix according to

$$\sigma_{b \leftarrow a}^{S\Gamma}(\epsilon) = \frac{\pi(2S+1)}{4\epsilon(2S_a+1)} \sum_{\alpha}^{a \in \alpha} \sum_{\beta}^{b \in \beta} |S_{\alpha\beta}^{S\Gamma}(E_a + \epsilon) - \delta_{\alpha\beta}|^2 \quad (19)$$

where $S_{\alpha\beta}^{S\Gamma}$ is the scattering matrix, S is the total spin, Γ denotes the molecular symmetry, ϵ is the incident electron energy, and $\delta_{\alpha\beta}$ is the Kronecker delta. Summing over all spin and symmetry contributions yields the total integral cross section,

$$\sigma_{b \leftarrow a}(\epsilon) = \sum_{S\Gamma} \sigma_{b \leftarrow a}^{S\Gamma}(\epsilon) \quad (20)$$

The integral cross section is the principal quantity investigated in this work. In particular, comparisons of the calculated cross sections with experimental measurements focus on reproducing characteristic low-energy scattering features, including the R-T minimum and resonance peaks, which provide sensitive tests of the accuracy of the underlying electronic structure and scattering models.

3 Methods

3.1 Computational Workflow

Methane, CH₄, was selected as the molecule of focus for this project because its low-energy electron scattering properties have been extensively investigated both experimentally and theoretically. In particular, the energies and magnitudes of the R-T minimum, resonance peaks, and electronic excitation energies have been reported in numerous studies, providing reliable reference data against which theoretical calculations may be compared. Consequently, CH₄ serves as an ideal test system for evaluating the accuracy of electron scattering methodologies. In addition to its well-characterized scattering properties, CH₄ is computationally attractive because it is a relatively small polyatomic molecule containing only ten electrons while still exhibiting molecular symmetry. Exploiting molecular symmetry significantly reduces the size of the electronic Hamiltonian and scattering matrices, thereby decreasing the computational expense of both electronic structure and close-coupling calculations without sacrificing accuracy. The computational workflow consisted of four major stages. First, the equilibrium geometry of methane was optimized using Hartree-Fock SCF calculations. Then, electron scattering calculations were carried out independently using

the UKRMol+ version 3.3.0 and ASTRA software packages. Systematic convergence studies were performed by varying the principal computational parameters, and finally, the resulting integral cross sections were compared with published theoretical and experimental data. Particular emphasis was placed on reproducing the R-T minimum and the low-energy resonance peak, as these provide sensitive benchmarks for assessing the quality of the electronic structure and scattering models.

3.2 Geometry Optimization

The equilibrium geometry of methane was obtained using Hartree-Fock SCF with the cc-pVTZ basis set. Geometry optimizations were performed using both the D_2 and C_{2v} symmetry representations of CH_4 . Although CH_4 belongs to the T_d point group, both UKRMol+ and ASTRA employ Abelian symmetry groups for computational efficiency.

3.3 UKRMol+ Scattering Calculations

Electron scattering calculations were performed using UKRMol+, an *ab initio* electron-molecule scattering package based on the close-coupling approximation and R-matrix formalism. The workflow for UKRMol+ is as follows: first molecular orbitals are generated by an external quantum chemistry package such as PSI4 or Molpro. The generated molecular orbitals are subsequently combined with continuum orbitals using the `scatci_integrals` program to evaluate the one- and two-electron integrals required for the scattering calculation. The `congen` program is then used to generate the configuration interaction expansions for both the N-electron target molecule and the corresponding (N+1)-electron scattering system. These Hamiltonian matrices are diagonalized using `scatci`, yielding the target energies and scattering-state eigenvectors required for the R-matrix calculation. Finally, the `rsolve` program propagates the R-matrix from the inner region to the asymptotic outer region, where the scattering K-matrix, T-matrix, and integral cross sections are obtained. Several scattering models of increasing sophistication were investigated. Static Exchange (SE) calculations employ a frozen Hartree-Fock target wavefunction and therefore included only exchange interactions between the incident electron and the target. Static Exchange plus Polarization (SEP) calculations additionally incorporate the long-range polarization interaction between the scattering electron and the target. Complete Active Space (CAS) and Multi-Reference Configuration Interaction (MRCI) calculations were also performed to provide a more accurate treatment of electron correlation and target polarization.

To test for convergence, cross section calculations were performed in which the R-matrix radius, maximum orbital angular momentum number (l_{max}), and number of target states included in the close-coupling expansion were systematically varied. Calculations were performed with R-matrix radii values of 10 a_0 , 13 a_0 , 15 a_0 , and 15 a_0 ; maximum orbital angular momentum numbers of 2, 3, and 4; and 2, 4, and 8 target states. Unless otherwise specified, UKRMol+ calculations employed the cc-pVTZ basis set and the C_{2v} symmetry representation.

3.4 ASTRA Scattering Calculations

In parallel, scattering calculations were carried out using ASTRA, which solves the electron scattering problem using TDMCC formalism rather than the R-matrix method. Within the ASTRA workflow, DALTON is first used to generate Hartree-Fock molecular orbitals and Gaussian basis functions. These orbitals are subsequently used by the LUCIA module to perform state-averaged multiconfiguration self-consistent field (MCSCF) and Complete Active Space Configuration Interaction (CASCI) calculations. LUCIA additionally computes the transition density matrices that couple the electronic target states during the scattering calculation. ASTRA then employs the PRISM hybrid integral library to evaluate the required molecular integrals before solving the coupled scattering equations and computing the scattering matrices and cross sections. Convergence studies for ASTRA focused primarily on the size of the active orbital space and the maximum continuum angular momentum. Active spaces containing 8 and 15 orbitals were investigated together with continuum angular momentum values of 3 and 4. ASTRA calculations employed the cc-pVQZ, aug-cc-pVQZ, and d-aug-cc-pVQZ basis sets.

3.5 Comparison with Literature

Following convergence testing, the highest-quality UKRMol+ and ASTRA calculations were compared with published theoretical and experimental integral cross sections for CH₄. The principal quantities used to assess the quality of the calculations were the position and depth of the R-T minimum, the energy and magnitude of the low-energy resonance peak, and the overall agreement of the elastic integral cross sections over the investigated energy range. As both the R-T minimum and resonance features are highly sensitive to the quality of the target electronic structure and the treatment of electron correlation and polarization, agreement with these benchmark quantities provides a stringent validation of the computational methodology.

4 Results and Discussion

4.1 Geometry Optimization

The optimized ground state energies of CH₄ obtained using the different software packages and molecular symmetry groups are summarized in Table 1. As shown in Table 1, the calculated ground state energies obtained with the C_{2v} and D_2 symmetry groups are in close agreement, differing by only 2×10^{-4} a.u. The UKRMol+ and ASTRA calculations performed with C_{2v} symmetry likewise produced energies in excellent agreement, demonstrating consistency between the two programs’ implementation of Hartree-Fock calculations. Since the C_{2v} representation was found to be the more commonly adopted symmetry for methane scattering calculations in the literature, it was selected for all subsequent calculations.

Software	Symmetry Group	Energy (a.u.)
ASTRA	C_{2v}	-40.2133
UKRMol+	C_{2v}	-40.2133
UKRMol+	D_2	-40.2135

Table 1: SCF Geometry optimization of CH_4 using the C_{2v} and D_2 symmetries.

To evaluate the quality of the electronic structure calculations, the target state energies computed with UKRMol+ and ASTRA were compared with benchmark values reported in the Computational Chemistry Comparison and Benchmark Database (CCCBDB) maintained by the National Institute of Standards and Technology (NIST). The UKRMol+ calculations employed the cc-pVTZ basis set, whereas the ASTRA calculations used the aug-cc-pVQZ basis set. Both calculations were performed with $l_{\text{max}} = 4$ and a CAS(8,8) active space. Table 2 summarizes the calculated ground and excited state energies together with the corresponding excitation energies. The NIST benchmark ground-state energies are -40.421 a.u. for the CID/cc-pVTZ calculation and -40.434 a.u. for the CID/aug-cc-pVQZ calculation. The UKRMol+ and ASTRA ground-state energies differ from their respective benchmark values by approximately 0.201 a.u. and 0.185 a.u., respectively, indicating good overall agreement with the reference calculations. In addition, the ASTRA target-state energies are consistently lower than the corresponding UKRMol+ values. This trend is consistent with the use of the larger augmented basis set as well as the fact that the target states generated by LUCIA have more closely correlated than the target states generated by UKRMol+. In addition, both approaches reproduced the expected energy ordering of the low-lying electronic states.

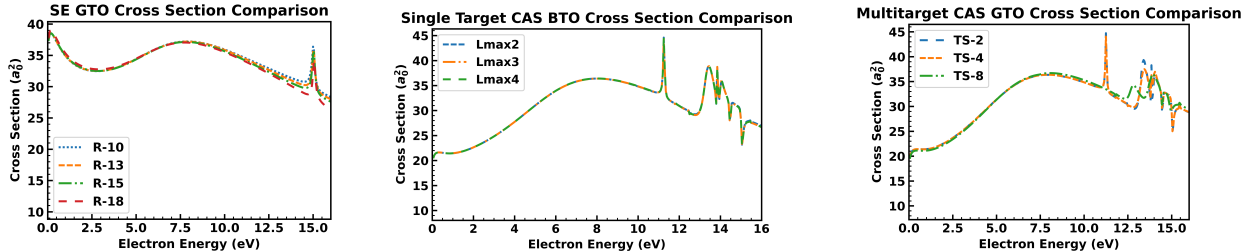
Target State	UKRMol+–CAS(8,8) (a.u.)	$\Delta E_{\text{UKRMol+}}$ (eV)	ASTRA–CAS(8,8) (a.u.)	ΔE_{ASTRA} (eV)
1A_1	-40.220	0.000	-40.249	0.000
$^3A_1 {}^3B_1 {}^3B_2$	-39.803	11.347	-39.890	9.769
$^1A_1 {}^1B_1 {}^1B_2$	-39.781	11.946	-39.878	10.095
3A_1	-39.755	12.653	-39.796	12.327

Table 2: Ground and excited state energies of methane computed using UKRMol+ and ASTRA with a CAS(8,8) active space. Excitation energies (ΔE) are given relative to the ground state.

4.2 UKRMol+ Scattering Calculations

Convergence testing was performed to determine appropriate computational parameters for the UKRMol+ scattering calculations. Three parameters were investigated: the R-matrix radius, the maximum orbital angular momentum, and the number of target states included in the close coupling expansion. The resulting integral cross sections are shown in Figure 1. Overall, the calculated cross sections exhibit only minor variations over the energy range of interest, indicating that the calculations are reasonably well converged below approximately 10 eV. Increasing the R-matrix radius from 10 a_0 to 18 a_0 produced negligible changes

in the calculated cross section, suggesting that the scattering wavefunction is adequately represented within the chosen R-matrix sphere. Likewise, increasing the maximum orbital angular momentum beyond 2 resulted in relatively small changes to the calculated cross sections, while increasing the number of target states from 2 to 8 primarily affected the detailed structure of the resonance region without significantly altering the overall scattering behavior.



(a) Convergence testing with R-matrix radii of $10 a_0$, $13 a_0$, $15 a_0$, and $18 a_0$. A SE calculation with GTO functions was performed for each value.

(b) Convergence testing with maximum orbital angular momentum values of 2, 3, and 4. An unpolarized CAS BTO calculation was performed for each value.

(c) Convergence testing with 2, 4, and 8 target states. A polarized CAS GTO calculation was performed for each value.

Figure 1: UKRMol+ convergence testing.

Figure 2 highlights the influence of l_{max} and the number of target states on the R-T minimum and resonance peak positions. Increasing the maximum orbital angular momentum shifted the calculated R-T minimum slightly, from approximately 1.3 eV to 1.2 eV. Although this represents an improvement, the calculated minimum remains considerably higher than the literature value of approximately 0.4 eV. This result suggests that, while the calculations are numerically converged with respect to these parameters, additional improvements to the target state description of basis set may be required to reproduce the experimentally-observed and theoretically-predicted R-T minimum more accurately. All UKRMol+ calculations presented in this section employed the cc-pVTZ basis set.

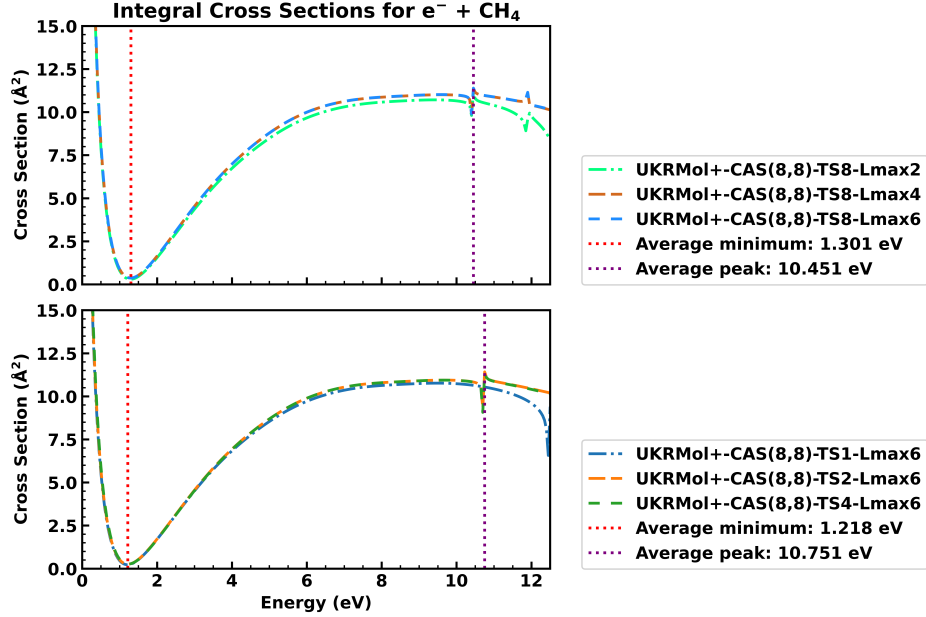


Figure 2: Convergence testing with $l_{max} \in \{3, 4\}$ and number of target states ranging from 2 to 8. After observing that the curves were well converged, the locations of the low-energy resonance peaks and R-T minimum were quantified using a Python script that identified the average energies at which the minimum and first peaks occurred.

4.3 ASTRA Scattering Calculations

Before performing new scattering calculations, the ASTRA method was first validated by reproducing a previously published static exchange calculation by Lima *et al* based on the Schwinger multichannel method. The comparison between the ASTRA calculation and the literature-provided values is shown in Figure 3. Excellent agreement was obtained between the calculated and published cross sections, with an R^2 value of 0.97. The ASTRA calculation employed the cc-pVTZ basis set with $l_{max} = 3$, and a single target state. The calculation produced a ground state energy of -40.213 a.u. The close agreement with the literature confirms that the ASTRA code is capable of accurately reproducing established scattering results.

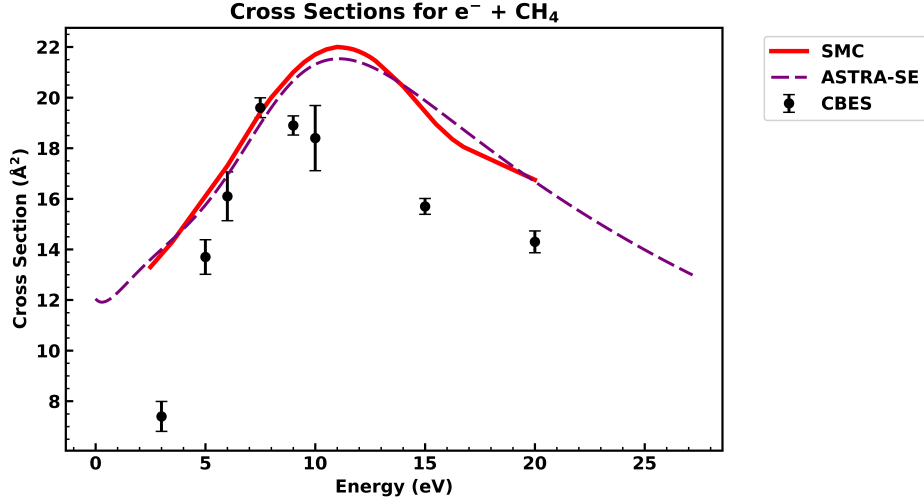
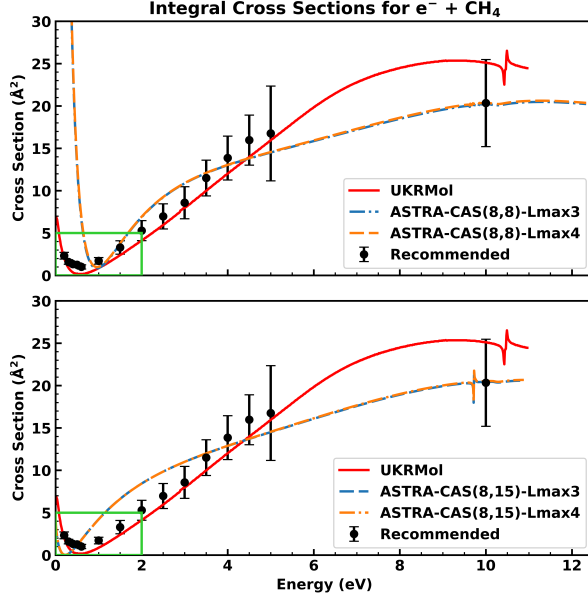
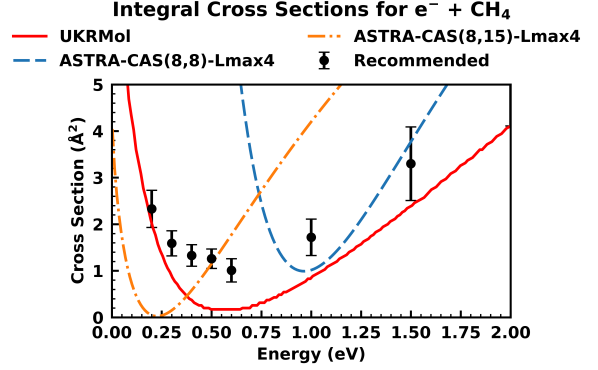


Figure 3: The ASTRA SE calculation was validated against the SE calculation conducted by Lima *et al*, who used the Schwinger multichannel method. Additionally, experimental data is included from cross beam electron spectrometry measurements collected by Tanaka *et al*.

Following validation, convergence testing was performed by varying both the maximum orbital angular momentum and the size of the active space. Calculations were performed for $l_{max} = 3$ and $l_{max} = 4$ using CAS(8,8) and CAS(8,15) active spaces. The resulting cross sections are shown in Figure 4. Increasing the active space from size 8 to size 15 significantly improved the description of the scattering process by allowing additional resonance features to emerge that were absent in the smaller active space calculations. Increasing the size of the active space also shifted the position of the R-T minimum from approximately 1.0 eV to 0.25 eV, bringing it much closer to the literature value of approximately 0.4 eV. All calculations used the aug-cc-pVQZ basis set and included 8 target states. The computed ground state energies were -40.249 a.u. for the CAS(8,8) calculations and -40.333 a.u. for the CAS(8,15) calculations, reflecting the improved treatment of electron correlation provided by the larger active space.



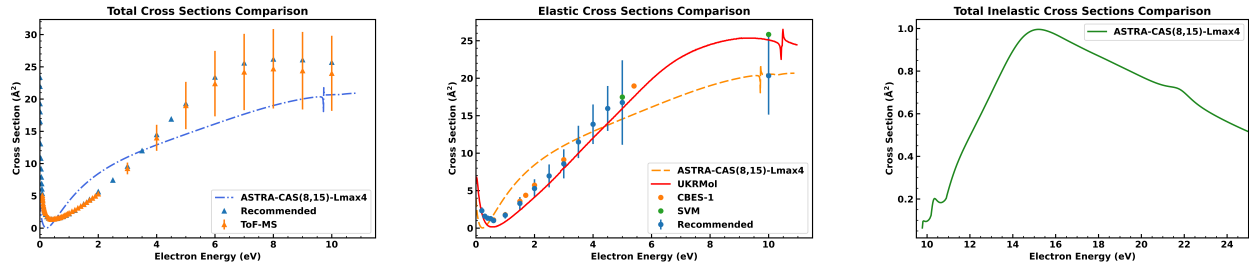
(a) Convergence testing with $l_{max} \in \{3, 4\}$ and active spaces of size 8 and 15.



(b) A closer view (green box in *Figure (a)*) of R-T minima computed and literature value.

Figure 4: ASTRA convergence testing.

Additionally, the total, elastic, and total inelastic cross sections obtained from the CAS(8,15) calculations are presented in Figure 5. The total inelastic cross section was obtained by summing the contributions from all non-elastic scattering channels. Overall, agreement was observed between the ASTRA total and elastic cross sections and the available theoretical and experimental literature data. These comparisons demonstrate that the combination of the larger active space and augmented basis set provides a reliable description of low-energy electron scattering from methane.



(a) Comparison of the total cross section computed with ASTRA with theoretical and experimental (ToF-MS: time of flight mass spectrometry by Ferch *et al*) literature values. Recommended values are compiled from theoretical and experimental sources and suggested by Song *et al*.

(b) Comparison of the elastic cross section computed with ASTRA with theoretical (UKRMol by Brigg *et al*, SVM: Schwinger variational method by Cho *et al*) and experimental (CBES: cross beam electron spectrometry by Bundschu *et al*) literature values.

(c) Total inelastic cross section computed by ASTRA.

Figure 5: ASTRA total, elastic, and total inelastic cross sections

4.4 Comparison with Literature

To evaluate the overall performance of the two scattering methods, the best performing calculations obtained with UKRMol+ and ASTRA were compared with the theoretical cross sections reported by Brigg *et al*. The UKRMol+ calculation employed a CAS(8,8) active space with eight target states, $l_{max} = 6$, and the cc-pVTZ basis set. The ASTRA calculation used a larger CAS(8,15) active space with $l_{max} = 4$ and the aug-cc-pVQZ basis set.

As shown in Figure 6, both methods successfully reproduced the principal features of the low-energy scattering cross section, including the R-T minimum and the low-energy resonance peak. However, noticeable differences were observed in the predicted positions of these features. As shown in Table 3, the ASTRA calculation predicted an R-T minimum at 0.232 eV, which is in much closer agreement with the literature value of approximately 0.4 eV than the UKRMol+ prediction of 1.301 eV. Similarly, the resonance peak predicted by ASTRA (9.727 eV) agrees more closely with the literature value than the corresponding UKRMol+ result (10.451 eV). The root-mean-square error (RMSE) of the ASTRA calculation is approximately three times smaller than that of the UKRMol+ calculation, indicating significantly better agreement with the reference cross section over the energy range considered. This improvement is likely attributable to the combination of the larger aug-cc-pVQZ basis set, the expanded CAS(8,15) active space, and the more accurate treatment of electron correlation in the target-state description. Nevertheless, both methods reproduce the overall shape of the cross section and correctly identify the major scattering features, demonstrating that each provides a physically meaningful description of low-energy electron scattering for methane.

	UKRMol+-CAS(8,8)-TS8-Lmax6	ASTRA-CAS(8,15)-Lmax4
Basis Set	cc-pVTZ	aug-cc-pVQZ
ΔE (eV)	11.341	9.776
R-T (eV)	1.301	0.232
Peak (eV)	10.451	9.727
RMSE (eV)	12.173	3.938

Table 3: .

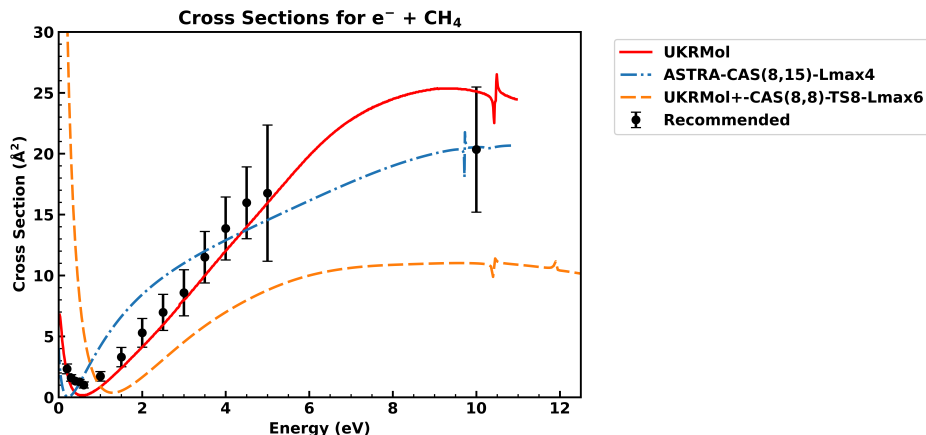


Figure 6: Comparison of best UKRMol+ and ASTRA calculations with the literature values provided by Brigg *et al.*

5 Conclusion

In this work, low-energy electron scattering from CH_4 was investigated using the UKRMol+ and ASTRA computational frameworks. Geometry optimization and target-state calculations demonstrated good agreement between the two methods and with benchmark electronic structure data, providing confidence in the molecular wavefunctions used for the scattering calculations.

Convergence studies were performed to determine suitable computational parameters for each approach. For UKRMol+, convergence with respect to the R-matrix radius, maximum orbital angular momentum, and number of target states was established. For ASTRA, the effects of the maximum orbital angular momentum and active-space size were investigated. Increasing the active-space size in the ASTRA calculations improved the description of electron correlation and resulted in a more accurate prediction of the R-T minimum and resonance features.

Comparison with published theoretical results showed that both UKRMol+ and ASTRA successfully reproduced the characteristic low-energy features of electron-methane scattering, including the R-T minimum and the broad shape resonance. Overall, the ASTRA calculations exhibited closer agreement with the literature, particularly in the positions of the R-T minimum and resonance peak, yielding a substantially lower root-mean-square error than

the corresponding UKRMol+ calculations. These differences likely arise from the use of a larger basis set, a more extensive active space, and differences in the treatment of electron correlation and the construction of the close-coupling expansion.

The methodologies developed in this study provide a foundation for future investigations of more complex molecular targets. In particular, they can be applied to tetrafluoromethane, CF_4 , an important per- and polyfluoroalkyl substance (PFAS) widely used in plasma-assisted semiconductor manufacturing. Extending these scattering calculations to CF_4 will contribute to a better understanding of electron-driven processes relevant to plasma chemistry and semiconductor fabrication.

6 References

1. Brigg *et al* *J. Phys. B.* **47**, 185203 (2014).
2. Bundschu *et al* *J. Phys. B.* **30**, 2239 (1997).
3. Cho *et al* *J. Phys. B.* **41**, 045203 (2008).
4. Ferch *et al* *J. Phys. B.* **18**, L445 (1985).
5. Houfek *et al* *Comput. Phys. Commun.* **298**, 109113 (2024).
6. Lima *et al* *Phys. Rev. A.* **32**, 2696–2701 (1985).
7. Lohmann *et al* *J. Phys. B.* **19**, 2565 (1986).
8. Song *et al* *J. Phys. Chem. Ref. Data* **44**, 023101 (2015).
9. Randazzo *et al* *Phys. Rev. Research* **5**, 043115 (2023).
10. Tanaka *et al* *J. Phys. B.* **15**, 3305–3319 (1982).
11. Ziółkowski *et al* *J. Chem. Phys.* **137**, 22A510 (2012).